

SOLIDWORKS API (.NET, VBA, C#) — Reference **(207/337): IModelView (part 1 of 2)**

SOLIDWORKS API Documentation (converted from CHM)

IModelView (part 1 of 2) (Type)

IModelView Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : IModelView Interface

Allows you access to the model view's orientation, translation, and the Microsoft handle to the window.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

Public Interface IModelView

Visual Basic (Usage)



Dim instance As [IModelView](#)

C#

public interface IModelView

C++/CLI

public interface class IModelView

Visual Basic for Applications (VBA) Syntax

See [ModelView](#).

Example

[Access Assembly \(C++\)](#)
[Get Scale Factor of Each Model View \(C++\)](#)
[Get Scale Factor of Each Model View \(C#\)](#)
[Get Scale Factor of Each Model View \(VB.NET\)](#)
[Get Scale Factor of Each Model View \(VBA\)](#)
[Create a Callout in a Model View \(C#\)](#)
[Create a Callout in a Model View \(VB.NET\)](#)
[Create a Callout in a Model View \(VBA\)](#)
[Get Names of Components, Window Handles, and DIBSECTIONS \(C#\)](#)
[Get Names of Components, Window Handles, and DIBSECTIONS \(VB.NET\)](#)
[Get Names of Components, Window Handles, and DIBSECTIONS \(VBA\)](#)

Remarks

Events are implemented with delegates in the Microsoft .NET Framework. See the [Overview](#) topic for a list of delegates for this interface.

Accessors

[IEnumModelViews::Next](#)
[IModelDoc2::ActiveView](#) and [IModelDoc2::IActiveView](#)
[IModelDoc2::GetFirstModelView](#) and [IModelDoc2::IGetFirstModelView](#)
[IModelDocExtension::GetModelViews](#) and [IModelDocExtension::IGetModelViews](#)
[IModelView::GetNext](#) and [IModelView::IGetNext](#)

Access Diagram

[ModelView](#)

See Also

IModelView Interface Members

[See Also](#) [Properties](#) [Methods](#)

[SolidWorks.Interop.sldworks Namespace](#) : IModelView Interface

The following tables list the members exposed by [IModelView](#).

Public Properties

Name	Description
 Camera	Gets or sets the camera.
 DisplayMode	Gets or sets the display mode for this model view.
 DisplayZebraStripes	Gets or sets the zebra-line display state.
 DynamicMode	Gets the dynamic mode.
 EnableGraphicsUpdate	Gets or sets whether or not to refresh the model view.
 FrameHeight	Get or sets the height of the frame of the document window that contains the model view in the SOLIDWORKS client area.
 FrameLeft	Gets or sets the left position of the frame of the document window that contains the model view in the SOLIDWORKS client area.
 FrameState	Gets or sets the window state (minimum, maximum, or normal) for the frame of the document window that contains the model view in the SOLIDWORKS client area.
 FrameTop	Gets or sets the top position of the frame of the document window that contains the model view in the SOLIDWORKS client area.
 FrameWidth	Gets or sets the width of the frame of the document window that contains the model view in the SOLIDWORKS client area.
 HirQuality	Gets or sets the hidden-line removal quality property of this model view.
 IOrientation	Obsolete. Superseded by IModelView.Orientation3 .
 IOrientation2	Obsolete. Superseded by IModelView.Orientation3 .
 ITranslation	Obsolete. Superseded by IModelView.Translation3 .
 ITranslation2	Obsolete. Superseded by IModelView.Translation3 .
 Linked	Gets whether or not this viewport is linked or not.
 ObjectSizesAway	Helps define the perspective of the current model view by relating the size of a displayed object with the distance of the object from the observer.
 Orientation	Obsolete. Superseded by IModelView.Orientation3 .
 Orientation2	Obsolete. Superseded by IModelView.Orientation3 .
 Orientation3	Gets or sets the model view orientation matrix.
 Scale	Obsolete. Superseded by IModelView.Scale2 .
 Scale2	Gets and sets the model view scale factor.
 SuppressWaitCursorDuringRedraw	Gets or sets the state of the wait cursor (also called the hourglass) while this model view is being redrawn.
 Transform	Gets the model space to the model view plane transform. NOTE: This property is a get-only property. Set is not implemented.
 Translation	Obsolete. Superseded by IModelView.Translation3 .
 Translation2	Obsolete. Superseded by IModelView.Translation3 .
 Translation3	Gets or sets the model view translation vector.
 UpAxisFlyout	Gets or sets the up axis flyout flag.
 UpdateAllGraphicsLayers	Gets or sets whether to update all graphic layers in any mode.
 VisibilityViewTools	Gets or sets the visibility of the Heads-up View Toolbar for this model view.
 XorHighlight	Gets or sets the Xor highlight state.

Public Methods

Name	Description
 Activate	Activates the model view.
 AddPerspective	Adds perspective to the model view.
 CreateCallout	Creates a callout on this model view.
 DrawHighlightedItems	Draws or redraws the highlighted items.
 DrawTransparentFeatureTree	Draws a transparent FeatureManager design tree.
 GetBkgdImageDisplayAreaAspectRatio	Gets the aspect ratio (width/height) of the area of the window where the background image is displayed.
 GetDisplayState	Gets the display state of this model view.
 GetEyePoint	Gets the eye position for perspective viewing.
 GetIsoPlaneDistance	Gets the distance, in terms of screen space, from the eye position to the plane at which there is no scaling due to perspective.
 GetMouse	Gets the mouse for this model view.
 GetNext	Gets the next model view after this model view.
 GetStereoSeparation	Obsolete and not superseded. Functionality no longer implemented.
 GetViewDIB	Obsolete. Superseded by IModelView::GetViewDIBx64. This method is unsafe to use with 64-bit SOLIDWORKS, i.e. all versions since SOLIDWORKS 2016. The 32-bit integer returned may be too small to contain the 64-bit window handle/pointer, and will be truncated silently. A critical exception will occur sometime later, when the truncated handle/pointer is used. Gets an image of the document as it looks in Normal view or in the print preview.
 GetViewDIBx64	Gets an image of the document as it looks in Normal view or in the print preview in 64-bit applications.
 GetViewHWnd	Obsolete. Superseded by IModelView::GetViewHWndx64. This method is unsafe to use with 64-bit SOLIDWORKS, i.e. all versions since SOLIDWORKS 2016. The 32-bit integer returned may be too small to contain the 64-bit window handle/pointer, and will be truncated silently. A critical exception will occur sometime later, when the truncated handle/pointer is used. Gets the Microsoft window handle for this model view.
 GetViewHWndx64	Gets the Microsoft window handle for this model view in 64-bit applications.
 GetViewPlaneDistance	Gets the model view plane distance for perspective viewing.
 GetVisibleBox	Gets the boundaries of the visible model view window.
 GraphicsRedraw	Redraws the specified region of or the entire SOLIDWORKS graphics window.
 HasPerspective	Determines if the model view currently has perspective.
 HideMagnifyingGlass	Hides the Magnifying Glass tool.
 IGetEyePoint	Gets the eye position for perspective viewing.
 IGetNext	Gets the next model view after this model view.
 IGetStereoSeparation	Obsolete and not superseded. Functionality no longer implemented.
 IGetVisibleBox	Gets the boundaries of the visible model view window.
 IGraphicsRedraw	Redraws the specified region of or the entire SOLIDWORKS graphics window.
 InitializeShading	Sets up the model view for OpenGL shading.
 ISetEyePoint	Sets the eye position for perspective viewing.
 ISetStereoSeparation	Obsolete and not superseded. Functionality no longer implemented.
 MoveMagnifyingGlass	Moves Magnifying Glass tool to the specified coordinates.
 RemovePerspective	Removes perspective from the model view.
 RollBy	Rolls the model view about the line of sight by the specified angle.
 RotateAboutAxis	Rotates the model view about a point, by an angle in the specified direction.

	RotateAboutCenter	Rotates the model view about the screen X and Y axes.
	RotateAboutPoint	Rotates the model view about the specified point by the specified angles in the directions of the screen X and Y axes.
	SetCameraByName	Sets the model view to that of the specified camera, if in camera view mode.
	SetEyePoint	Sets the eye position for perspective viewing.
	SetStereoSeparation	Obsolete and not superseded. Functionality no longer implemented.
	ShowMagnifyingGlass	Shows the Magnifying Glass tool at the specified coordinates.
	StartDynamics	Provides faster rotation of model views.
	StopDynamics	Ends dynamic model view motion.
	TranslateBy	Translates the model view in the screen.
	TurnBy	Rotates the camera by the specified angles about the camera's x and y axes.
	ZoomByFactor	Modifies the zoom factor for the model view.

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See Also

[IModelView Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IModelViewManager Interface](#)
[IModelWindow Interface](#)

SOLIDWORKS API Help

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IModelView Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IModelView Interface

For a list of all members of this type, see [IModelView members](#).

Public Methods

Name	Description
	Activate Activates the model view.
	AddPerspective Adds perspective to the model view.
	CreateCallout Creates a callout on this model view.
	DrawHighlightedItems Draws or redraws the highlighted items.
	DrawTransparentFeatureTree Draws a transparent FeatureManager design tree.
	GetBkgdImageDisplayAreaAspectRatio Gets the aspect ratio (width/height) of the area of the window where the background image is displayed.
	GetDisplayState Gets the display state of this model view.
	GetEyePoint Gets the eye position for perspective viewing.
	GetIsoPlaneDistance Gets the distance, in terms of screen space, from the eye position to the plane at which there is no scaling due to perspective.
	GetMouse Gets the mouse for this model view.
	GetNext Gets the next model view after this model view.
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The 32-bit integer returned may be too small to contain the 64-bit window handle/pointer, and will be truncated silently. A critical exception will occur sometime later, when the truncated handle/pointer is used.

Gets the Microsoft window handle for this model view.

	GetViewHWnd64	Gets the Microsoft window handle for this model view in 64-bit applications.
	GetViewPlaneDistance	Gets the model view plane distance for perspective viewing.
	GetVisibleBox	Gets the boundaries of the visible model view window.
	GraphicsRedraw	Redraws the specified region of or the entire SOLIDWORKS graphics window.
	HasPerspective	Determines if the model view currently has perspective.
	HideMagnifyingGlass	Hides the Magnifying Glass tool.
	IGetEyePoint	Gets the eye position for perspective viewing.
	IGetNext	Gets the next model view after this model view.
	IGetStereoSeparation	Obsolete and not superseded. Functionality no longer implemented.
	IGetVisibleBox	Gets the boundaries of the visible model view window.
	IGraphicsRedraw	Redraws the specified region of or the entire SOLIDWORKS graphics window.
	InitializeShading	Sets up the model view for OpenGL shading.
	ISetEyePoint	Sets the eye position for perspective viewing.
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	MoveMagnifyingGlass	Moves Magnifying Glass tool to the specified coordinates.
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 See Also

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IModelView Interface Properties

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IModelView Interface

For a list of all members of this type, see [IModelView members](#).

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 FrameHeight	Get or sets the height of the frame of the document window that contains the model view in the SOLIDWORKS client area.
 FrameLeft	Gets or sets the left position of the frame of the document window that contains the model view in the SOLIDWORKS client area.
 FrameState	Gets or sets the window state (minimum, maximum, or normal) for the frame of the document window that contains the model view in the SOLIDWORKS client area.
 FrameTop	Gets or sets the top position of the frame of the document window that contains the model view in the SOLIDWORKS client area.
 FrameWidth	Gets or sets the width of the frame of the document window that contains the model view in the SOLIDWORKS client area.
 HlrQuality	Gets or sets the hidden-line removal quality property of this model view.
 IOrientation	Obsolete. Superseded by IModelView.Orientation3 .
 IOrientation2	Obsolete. Superseded by IModelView.Orientation3 .
 ITranslation	Obsolete. Superseded by IModelView.Translation3 .
 ITranslation2	Obsolete. Superseded by IModelView.Translation3 .
 Linked	Gets whether or not this viewport is linked or not.
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 XorHighlight	Gets or sets the Xor highlight state.

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[IModelViewManager Interface](#)
[IModelWindow Interface](#)

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Activate Method (IModelView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : Activate Method (IModelView)

Activates the model view.

.NET Syntax

Visual Basic (Declaration)

```
Sub Activate()
```

Visual Basic (Usage)



```
Dim instance As IModelView  
  
instance.Activate()
```

C#

```
void Activate()
```

C++/CLI

```
void Activate();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelView::Activate](#).

See Also

[IModelView Interface](#)
[IModelView Members](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

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AddPerspective Method (IModelView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : AddPerspective Method (IModelView)

Adds perspective to the model view.

.NET Syntax

Visual Basic (Declaration)

```
Sub AddPerspective()
```

Visual Basic (Usage)



```
Dim instance As IModelView  
  
instance.AddPerspective()
```

C#

```
void AddPerspective()
```

C++/CLI

```
void AddPerspective();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelView::AddPerspective](#).

See Also

[IModelView Interface](#)
[IModelView Members](#)
[IModelView::GetEyePoint Method](#)
[IModelView::!GetEyePoint Method](#)
[IModelView::!SetEyePoint Method](#)
[IModelView::SetEyePoint Method](#)
[IModelView::HasPerspective Method](#)
[IModelView::GetViewPlaneDistance Method](#)
[IModelView::RemovePerspective Method](#)
[IModelView::ObjectSizesAway Property](#)

SOLIDWORKS API Help

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Camera Property (IModelView)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : Camera Property (IModelView)

Gets or sets the camera.

.NET Syntax

Visual Basic (Declaration)

Property Camera As Camera

Visual Basic (Usage)



```
Dim instance As IModelView  
Dim value As Camera
```

```
instance.Camera = value
```

```
value = instance.Camera
```

C#

```
Camera Camera {get; set;}
```

C++/CLI

```
property Camera^ Camera {  
    Camera^ get();  
    void set ( & Camera^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

[Camera](#)

Visual Basic for Applications (VBA) Syntax

See [ModelView::Camera](#).

Example

[Turn Cameras On and Off \(VBA\)](#)

Remarks

If the return value is Nothing or NULL, then the camera view for this model is turned off.

See Also

[IModelView Interface](#)
[IModelView Members](#)
[IModelView::SetCameraByName Method](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

SOLIDWORKS API Help

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CreateCallout Method (IModelView)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : CreateCallout Method (IModelView)

Creates a callout on this model view.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateCallout( _  
    ByVal NumberOfRows As System.Integer, _  
    ByVal LpHandler As System.Object _  
) As Callout
```

Visual Basic (Usage)



```
Dim instance As IModelView  
Dim NumberOfRows As System.Integer  
Dim LpHandler As System.Object  
Dim value As Callout  
  
value = instance.CreateCallout(NumberOfRows, LpHandler)
```

C#

```
Callout CreateCallout(  
    System.int NumberOfRows,  
    System.object LpHandler  
)
```

C++/CLI

```
Callout^ CreateCallout(  
& System.int NumberOfRows,  
& System.Object^ LpHandler  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NumberOfRows
Number of rows in the callout
LpHandler
Pointer to [ISwCalloutHandler](#)

Return Value

An [ICallout](#) object

Visual Basic for Applications (VBA) Syntax

See [ModelView::CreateCallout](#).

Example

[Create a Callout in a Model View \(VBA\)](#)
[Create a Callout in a Model View \(VB.NET\)](#)
[Create a Callout in a Model View \(C#\)](#)

See Also

[IModelView Interface](#)
[IModelView Members](#)

Availability

SOLIDWORKS 2011 FCS, Revision Number 19.0

SOLIDWORKS API Help

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DisplayMode Property (IModelView)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : DisplayMode Property (IModelView)

Gets or sets the display mode for this model view.

.NET Syntax

Visual Basic (Declaration)

Property DisplayMode As System.Integer

Visual Basic (Usage)



```
Dim instance As IModelView
Dim value As System.Integer

instance.DisplayMode = value
value = instance.DisplayMode
```

C#

```
System.int DisplayMode {get; set;}
```

C++/CLI

```
property System.int DisplayMode {
    System.int get();
    void set ( & System.int value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Display mode as defined by [swViewDisplayMode_e](#)

Visual Basic for Applications (VBA) Syntax

See [ModelView::DisplayMode](#).

Example

[Set Model Display Mode \(VBA\)](#)

See Also

[IModelView Interface](#)
[IModelView Members](#)
[IModelView::StartDynamics Method](#)
[IModelView::StopDynamics Method](#)
[IModelView::HlrQuality Property](#)
[IModelView::DynamicMode Property](#)

Availability

SOLIDWORKS 2004 SP1, Revision Number 12.1

SOLIDWORKS API Help

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DisplayZebraStripes Property (IModelView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : DisplayZebraStripes Property (IModelView)

Gets or sets the zebra-line display state.

 .NET Syntax

Visual Basic (Declaration)

Property DisplayZebraStripes As System.Boolean

Visual Basic (Usage)

Dim instance As [IModelView](#)
Dim value As System.Boolean

instance.DisplayZebraStripes = value

value = instance.DisplayZebraStripes

C#

System.bool DisplayZebraStripes {get; set;}

C++/CLI

property System.bool DisplayZebraStripes {
 System.bool get();
 void set (& System.bool value);
}

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to display the zebra stripes, false if not

 Visual Basic for Applications (VBA) Syntax

See [ModelView::DisplayZebraStripes](#).

 See Also

[IModelView Interface](#)
[IModelView Members](#)

 Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

DrawHighlightedItems Method (IModelView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : DrawHighlightedItems Method (IModelView)

Draws or redraws the highlighted items.

 .NET Syntax

Visual Basic (Declaration)

Sub DrawHighlightedItems()

Visual Basic (Usage)

Dim instance As [IModelView](#)

instance.DrawHighlightedItems()

C#

```
void DrawHighlightedItems()
```

C++/CLI

```
void DrawHighlightedItems();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelView::DrawHighlightedItems](#).

Remarks

This method is useful in applications that handle repainting if you want highlighting to be visible. When called, the currently highlighted items are painted.

See Also

[IModelView Interface](#)
[IModelView Members](#)

SOLIDWORKS API Help

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DrawTransparentFeatureTree Method (IModelView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : DrawTransparentFeatureTree Method (IModelView)

Draws a transparent FeatureManager design tree.

.NET Syntax

Visual Basic (Declaration)

```
Sub DrawTransparentFeatureTree()
```

Visual Basic (Usage)



```
Dim instance As IModelView  
instance.DrawTransparentFeatureTree()
```

C#

```
void DrawTransparentFeatureTree()
```

C++/CLI

```
void DrawTransparentFeatureTree();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelView::DrawTransparentFeatureTree](#).

Remarks

You can use this method to redraw the flyout FeatureManager design tree after rendering a model.

See Also

[IModelView Interface](#)
[IModelView Members](#)
[IModelDoc2::GraphicsRedraw2 Method](#)
[IModelDoc2::WindowRedraw Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

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DynamicMode Property (IModelView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : DynamicMode Property (IModelView)

Gets the dynamic mode.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property DynamicMode As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModelView
Dim value As System.Integer

value = instance.DynamicMode
```

C#

```
System.int DynamicMode {get;}
```

C++/CLI

```
property System.int DynamicMode {
    System.int get();
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Dynamic mode state as defined in [swDynamicMode_e](#)

Visual Basic for Applications (VBA) Syntax

See [ModelView::DynamicMode](#).

Remarks

If there is no geometry in the part or assembly, then this method returns swNoDynamics regardless of whether spinning, panning, zooming.

See Also

[IModelView Interface](#)
[IModelView Members](#)
[IModelView::StartDynamics Method](#)
[IModelView::StopDynamics Method](#)
[IModelView::HlrQuality Property](#)

SOLIDWORKS API Help

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EnableGraphicsUpdate Property (IModelView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : EnableGraphicsUpdate Property (IModelView)

Gets or sets whether or not to refresh the model view.

.NET Syntax

Visual Basic (Declaration)

```
Property EnableGraphicsUpdate As System.Boolean
```

```
Dim instance As IModelView
Dim value As System.Boolean

instance.EnableGraphicsUpdate = value

value = instance.EnableGraphicsUpdate
```

C#

```
System.bool EnableGraphicsUpdate {get; set;}
```

C++/CLI

```
property System.bool EnableGraphicsUpdate {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to refresh the model view, false to not

Visual Basic for Applications (VBA) Syntax

See [ModelView::EnableGraphicsUpdate](#).

Remarks

This property affects whether to refresh the model view during a selection, such as [IEntity::Select4](#) or [IFeature::Select2](#).

See Also

[IModelView Interface](#)
[IModelView Members](#)

Availability

SOLIDWORKS 2004 SP3, Revision Number 12.3

SOLIDWORKS API Help

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FrameHeight Property (IModelView)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : FrameHeight Property (IModelView)

Get or sets the height of the frame of the document window that contains the model view in the SOLIDWORKS client area.

.NET Syntax

Visual Basic (Declaration)

```
Property FrameHeight As System.Integer
```

Visual Basic (Usage)

Copy Code

```
Dim instance As IModelView
Dim value As System.Integer

instance.FrameHeight = value

value = instance.FrameHeight
```

C#

```
System.int FrameHeight {get; set;}
```

C++/CLI

```
property System.int FrameHeight {  
    System.int get();  
    void set ( & System.int value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Height of document window in client area in pixels

Visual Basic for Applications (VBA) Syntax

See [ModelView::FrameHeight](#).

Example

[Position Document Window \(VBA\)](#)

See Also

[IModelView Interface](#)
[IModelView Members](#)
[IModelView::FrameLeft Property](#)
[IModelView::FrameState Property](#)
[IModelView::FrameTop Property](#)
[IModelView::FrameWidth Property](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

FrameLeft Property (IModelView)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : FrameLeft Property (IModelView)

Gets or sets the left position of the frame of the document window that contains the model view in the SOLIDWORKS client area.

.NET Syntax

Visual Basic (Declaration)

```
Property FrameLeft As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModelView  
Dim value As System.Integer  
  
instance.FrameLeft = value  
value = instance.FrameLeft
```

C#

```
System.int FrameLeft {get; set;}
```

C++/CLI

```
property System.int FrameLeft {  
    System.int get();  
    void set ( & System.int value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Left position of document window in client area in pixels

Visual Basic for Applications (VBA) Syntax

See [ModelView::FrameLeft](#).

Example

[Position Document Window \(VBA\)](#)

See Also

[IModelView Interface](#)
[IModelView Members](#)
[IModelView::FrameHeight Property](#)
[IModelView::FrameState Property](#)
[IModelView::FrameTop Property](#)
[IModelView::FrameWidth Property](#)

SOLIDWORKS API Help

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FrameState Property (IModelView)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : FrameState Property (IModelView)

Gets or sets the window state (minimum, maximum, or normal) for the frame of the document window that contains the model view in the SOLIDWORKS client area.

.NET Syntax

Visual Basic (Declaration)

```
Property FrameState As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModelView
Dim value As System.Integer

instance.FrameState = value

value = instance.FrameState
```

C#

```
System.int FrameState {get; set;}
```

C++/CLI

```
property System.int FrameState {
    System.int get();
    void set ( & System.int value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Window state as defined in [swWindowState](#) e

Visual Basic for Applications (VBA) Syntax

See [ModelView::FrameState](#).

Example

[Position Document Window \(VBA\)](#)
[Create Hidden Undo Object \(VBA\)](#)
[Create Hidden Undo Object \(VB.NET\)](#)
[Create Hidden Undo Object \(C#\)](#)

See Also

[IModelView Interface](#)
[IModelView Members](#)
[IModelView::FrameHeight Property](#)
[IModelView::FrameLeft Property](#)
[IModelView::FrameTop Property](#)
[IModelView::FrameWidth Property](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

FrameTop Property (IModelView)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : FrameTop Property (IModelView)

Gets or sets the top position of the frame of the document window that contains the model view in the SOLIDWORKS client area.

.NET Syntax

Visual Basic (Declaration)

Property FrameTop As System.Integer

Visual Basic (Usage)



```
Dim instance As IModelView  
Dim value As System.Integer  
  
instance.FrameTop = value  
  
value = instance.FrameTop
```

C#

```
System.int FrameTop {get; set;}
```

C++/CLI

```
property System.int FrameTop {  
    System.int get();  
    void set ( & System.int value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Top position of document window in client area in pixels

Visual Basic for Applications (VBA) Syntax

See [ModelView::FrameTop](#).

Example

[Position Document Window \(VBA\)](#)

See Also

[IModelView Interface](#)
[IModelView Members](#)
[IModelView::FrameHeight Property](#)
[IModelView::FrameLeft Property](#)
[IModelView::FrameState Property](#)
[IModelView::FrameWidth Property](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

FrameWidth Property (IModelView)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : FrameWidth Property (IModelView)

Gets or sets the width of the frame of the document window that contains the model view in the SOLIDWORKS client area.

.NET Syntax

```
Property FrameWidth As System.Integer
```



```
Dim instance As IModelView
Dim value As System.Integer

instance.FrameWidth = value

value = instance.FrameWidth
```

```
System.int FrameWidth {get; set;}
```

```
property System.int FrameWidth {
    System.int get();
    void set ( & System.int value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Width of document window in client area in pixels

Visual Basic for Applications (VBA) Syntax

See [ModelView::FrameWidth](#).

Example

[Position Document Window \(VBA\)](#)

See Also

[IModelView Interface](#)
[IModelView Members](#)
[IModelView::FrameHeight Property](#)
[IModelView::FrameLeft Property](#)
[IModelView::FrameState Property](#)
[IModelView::FrameTop Property](#)

GetBkgdImageDisplayAreaAspectRatio Method (IModelView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : GetBkgdImageDisplayAreaAspectRatio Method (IModelView)

Gets the aspect ratio (width/height) of the area of the window where the background image is displayed.

.NET Syntax

```
Function GetBkgdImageDisplayAreaAspectRatio() As System.Double
```



```
Dim instance As IModelView
Dim value As System.Double

value = instance.GetBkgdImageDisplayAreaAspectRatio()
```

```
System.double GetBkgdImageDisplayAreaAspectRatio()
```

C++/CLI

```
System.double GetBkgdImageDisplayAreaAspectRatio();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Aspect ratio (width/height) of the area of the window where the background image is displayed, or -1 if there is no background image or if the image covers the entire window

Visual Basic for Applications (VBA) Syntax

See [ModelView::GetBkgdImageDisplayAreaAspectRatio](#).

Remarks

For use with ray-trace rendering engines to match SOLIDWORKS' display.

See Also

[IModelView Interface](#)
[IModelView Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetDisplayState Method (IModelView)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : GetDisplayState Method (IModelView)

Gets the display state of this model view.

.NET Syntax

Visual Basic (Declaration)

```
Function GetDisplayState( _  
    ByVal DisplayType As System.Integer _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelView  
Dim DisplayType As System.Integer  
Dim value As System.Boolean  
  
value = instance.GetDisplayState(DisplayType)
```

C#

```
System.bool GetDisplayState(  
    System.int DisplayType  
)
```

C++/CLI

```
System.bool GetDisplayState(  
& System.int DisplayType  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

DisplayType

Display setting to check as defined in [swViewDisplayType_e](#)

Return Value

True if the setting specified is turned on, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelView::GetDisplayState](#).

Example

[Get Display State \(VBA\)](#)

Remarks

By passing in an available DisplayType option, you can check various display conditions for the view. For example, if you want to know if a view is shaded, you could make the following call in Visual Basic:

```
res = mView.GetDisplayState( swIsViewShaded )
```

See Also

[IModelView Interface](#)
[IModelView Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetEyePoint Method (IModelView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : GetEyePoint Method (IModelView)

Gets the eye position for perspective viewing.

.NET Syntax

Visual Basic (Declaration)

```
Function GetEyePoint() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelView
Dim value As System.Object

value = instance.GetEyePoint()
```

C#

```
System.Object GetEyePoint()
```

C++/CLI

```
System.Object^ GetEyePoint();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of 3 doubles describing the eye position in screen space

Visual Basic for Applications (VBA) Syntax

See [ModelView::GetEyePoint](#).

Remarks

The values are returned in pixels and are the location of the eyepoint relative to the screen space origin (upper-left corner of window). The Z value reflects the distance from the eye point to the object center. These values are affected by the view scale.

To use these values, you should apply them after you have obtained the current view orientation matrix using ModelView::Xform.

The Z value (in pixels) will typically be close to

```
[ ( IModelView::GetViewPlaneDistance / ("Object Sizes Away") ) * ("Object Sizes Away" + 0.5) ]
```

The perspective view is created based on the value entered by the end-user for **Object Sizes Away**. If the user specifies **3 objects away**, then the eye point is positioned 3 body diameters from the view plane, where the view plane is located at the front of the body as seen from this particular orientation.

See Also

[IModelView Interface](#)
[IModelView Members](#)
[IModelView::AddPerspective Method](#)
[IModelView::!GetEyePoint Method](#)
[IModelView::!SetEyePoint Method](#)
[IModelView::SetEyePoint Method](#)
[IModelView::GetViewPlaneDistance Method](#)
[IModelView::HasPerspective Method](#)
[IModelView::RemovePerspective Method](#)
[IModelView::Scale2 Property](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetIsoPlaneDistance Method (IModelView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : GetIsoPlaneDistance Method (IModelView)

Gets the distance, in terms of screen space, from the eye position to the plane at which there is no scaling due to perspective.

.NET Syntax

Visual Basic (Declaration)

```
Function GetIsoPlaneDistance() As System.Double
```

Visual Basic (Usage)



```
Dim instance As IModelView  
Dim value As System.Double
```

```
value = instance.GetIsoPlaneDistance()
```

C#

```
System.double GetIsoPlaneDistance()
```

C++/CLI

```
System.double GetIsoPlaneDistance();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Distance

Visual Basic for Applications (VBA) Syntax

See [ModelView::GetIsoPlaneDistance](#).

See Also

[IModelView Interface](#)
[IModelView Members](#)
[IModelView::AddPerspective Method](#)
[IModelView::GetEyePoint Method](#)
[IModelView::GetViewPlaneDistance Method](#)
[IModelView::!GetEyePoint Method](#)
[IModelView::!SetEyePoint Method](#)
[IModelView::SetEyePoint Method](#)
[IModelView::HasPerspective Method](#)
[IModelView::RemovePerspective Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetMouse Method (IModelView)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : GetMouse Method (IModelView)

Gets the mouse for this model view.

.NET Syntax

Visual Basic (Declaration)

```
Function GetMouse() As Mouse
```

Visual Basic (Usage)



```
Dim instance As IModelView  
Dim value As Mouse  
  
value = instance.GetMouse()
```

C#

```
Mouse GetMouse()
```

C++/CLI

```
Mouse^ GetMouse();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[Mouse](#)

Visual Basic for Applications (VBA) Syntax

See [ModelView::GetMouse](#).

Example

[Run SOLIDWORKS Commands and Synthesize Mouse Events \(C#\)](#)
[Run SOLIDWORKS Commands and Synthesize Mouse Events \(VB.NET\)](#)
[Run SOLIDWORKS Commands and Synthesize Mouse Events \(VBA\)](#)

See Also

[IModelView Interface](#)
[IModelView Members](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetNext Method (IModelView)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : GetNext Method (IModelView)

Gets the next model view after this model view.

.NET Syntax

Visual Basic (Declaration)

```
Function GetNext() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelView  
Dim value As System.Object  
  
value = instance.GetNext()
```

C#

```
System.object GetNext()
```

C++/CLI

```
System.Object^ GetNext();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Next [model view](#) or Nothing or null if no more model views exist

Visual Basic for Applications (VBA) Syntax

See [ModelView::GetNext](#).

Example

[Get Names of Components, Window Handles, and DIBSECTIONS \(VBA\)](#)
[Get Names of Components, Window Handles, and DIBSECTIONS \(C#\)](#)
[Get Names of Components, Window Handles, and DIBSECTIONS \(VB.NET\)](#)

Remarks

You can traverse through the model views in a document by using this method. When this method returns Nothing or null, you have reached the end of the list.

See [IModelDoc2::EnumModelViews](#) for a method for traversing the model views on a document.

See Also

[IModelView Interface](#)
[IModelView Members](#)
[IEnumModelViews::Next Method](#)
[IModelDoc2::ActiveView Property](#)
[IModelDoc2::IActiveView Property](#)
[IModelDoc2::IGetFirstModelView Method](#)
[IModelDoc2::GetFirstModelView Method](#)
[IModelView::IGetNext Method](#)

Availability

SOLIDWORKS 98Plus, datecode 1998319

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetStereoSeparation Method (IModelView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : GetStereoSeparation Method (IModelView)

Obsolete and not superseded. Functionality no longer implemented.

.NET Syntax

Visual Basic (Declaration)

```
Function GetStereoSeparation() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelView  
Dim value As System.Object
```

```
value = instance.GetStereoSeparation()
```

C#

```
System.object GetStereoSeparation()
```

C++/CLI

```
System.Object^ GetStereoSeparation();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of 2 doubles representing the stereo magnitude and stereo parallax balance settings

Visual Basic for Applications (VBA) Syntax

See [ModelView::GetStereoSeparation](#).

Remarks

This method checks the two stereoscopic display values which can be set by using [IModelView::SetStereoSeparation](#) or [IModelView::ISetStereoSeparation](#).

See Also

[IModelView Interface](#)
[IModelView Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetViewDIB Method (IModelView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : GetViewDIB Method (IModelView)

Obsolete. Superseded by [IModelView::GetViewDIBx64](#).

This method is unsafe to use with 64-bit SOLIDWORKS, i.e. all versions since SOLIDWORKS 2016.

The 32-bit integer returned may be too small to contain the 64-bit window handle/pointer, and will be truncated silently. A critical exception will occur sometime later, when the truncated handle/pointer is used.

Gets an image of the document as it looks in Normal view or in the print preview.

.NET Syntax

Visual Basic (Declaration)

```
Function GetViewDIB() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModelView
Dim value As System.Integer

value = instance.GetViewDIB()
```

C#

```
System.int GetViewDIB()
```

C++/CLI

```
System.int GetViewDIB();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Pointer to DIBSECTION for image

Visual Basic for Applications (VBA) Syntax

See [ModelView::GetViewDIB](#).

Remarks

All applications are required to be compatible with SOLIDWORKS x64. You must replace all calls to 32-bit only APIs with calls to their 64-bit safe counterparts.

See [Eliminating 32-bit only API use](#) for more information.

The image type and resolution can be controlled interactively under TIFF export options.

If TIFF export is set to screen, then the output from this method is based on the current screen resolution. If it's set to print, then the output is based on the DPI specified in the TIFF options dialog.

The memory for the image bits (DIBSECTION.dsBm.bmBits) and this structure are allocated by the SOLIDWORKS Design application and must be deleted by the caller. For more information, see the description of Bitmap structures and DIBSECTION in Microsoft Help.

See Also

[IModelView Interface](#)
[IModelView Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetViewDIBx64 Method (IModelView)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : GetViewDIBx64 Method (IModelView)

Gets an image of the document as it looks in Normal view or in the print preview in 64-bit applications.

.NET Syntax

Visual Basic (Declaration)

```
Function GetViewDIBx64() As System.Long
```

Visual Basic (Usage)



```
Dim instance As IModelView
Dim value As System.Long

value = instance.GetViewDIBx64()
```

C#

```
System.long GetViewDIBx64()
```

C++/CLI

```
System.int64 GetViewDIBx64();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Pointer to DIBSECTION for image

Visual Basic for Applications (VBA) Syntax

See [ModelView::GetViewDIBx64](#).

Example

[Get Names of Components, Window Handles, and DIBSECTIONS \(C#\)](#)
[Get Names of Components, Window Handles, and DIBSECTIONS \(VB.NET\)](#)
[Get Names of Components, Window Handles, and DIBSECTIONS \(VBA\)](#)

Remarks

The image type and resolution can be controlled interactively under TIFF export options.

If TIFF export is set to screen, then the output from this method is based on the current screen resolution. If it's set to print, then the output is based on the DPI specified in the TIFF options dialog.

The memory for the image bits (DIBSECTION.dsBm.bmBits) and this structure are allocated by the SOLIDWORKS Design application and must be deleted by the caller. For more information, see the description of Bitmap structures and DIBSECTION in Microsoft Help.

NOTE: This method is only available through early binding and with 64-bit versions of the SOLIDWORKS Design software.

See Also

[IModelView Interface](#)
[IModelView Members](#)
[IModeler::GetViewDIB Method](#)

Availability

SOLIDWORKS 2006 SP2, Revision Number 14.2

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetViewHWnd Method (IModelView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : GetViewHWnd Method (IModelView)

Obsolete. Superseded by [IModelView::GetViewHWndx64](#).

This method is unsafe to use with 64-bit SOLIDWORKS, i.e. all versions since SOLIDWORKS 2016.

The 32-bit integer returned may be too small to contain the 64-bit window handle/pointer, and will be truncated silently. A critical exception will occur sometime later, when the truncated handle/pointer is used.

Gets the Microsoft window handle for this model view.

.NET Syntax

Visual Basic (Declaration)

```
Function GetViewHWnd() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModelView  
Dim value As System.Integer  
  
value = instance.GetViewHWnd()
```

C#

```
System.int GetViewHWnd()
```

C++/CLI

```
System.int GetViewHWnd();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Microsoft window handle

Visual Basic for Applications (VBA) Syntax

See [ModelView::GetViewHWnd](#).

Remarks

All applications are required to be compatible with SOLIDWORKS x64. You must replace all calls to 32-bit only APIs with calls to their 64-bit safe counterparts.

See [Eliminating 32-bit only API use](#) for more information.

Window handles are not unique across computers.

See Also

[IModelView Interface](#)
[IModelView Members](#)
[IModelWindow::HWnd Property](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetViewHWndx64 Method (IModelView)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : GetViewHWndx64 Method (IModelView)

Gets the Microsoft window handle for this model view in 64-bit applications.

.NET Syntax

Visual Basic (Declaration)

```
Function GetViewHWndx64() As System.Long
```

Visual Basic (Usage)



```
Dim instance As IModelView  
Dim value As System.Long
```

```
value = instance.GetViewHWndx64()
```

C#

```
System.long GetViewHWndx64()
```

C++/CLI

```
System.int64 GetViewHWndx64();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Microsoft window handle

Visual Basic for Applications (VBA) Syntax

See [ModelView::GetViewHWndx64](#).

Example

[Get Names of Components, Window Handles, and DIBSECTIONS \(C#\)](#)
[Get Names of Components, Window Handles, and DIBSECTIONS \(VB.NET\)](#)
[Get Names of Components, Window Handles, and DIBSECTIONS \(VBA\)](#)

Remarks

This method is only available through early binding and with 64-bit versions of the SOLIDWORKS Design software.
Window handles are not unique across computers.

See Also

[IModelView Interface](#)
[IModelView Members](#)
[IModelWindow::HWnd Property](#)

Availability

SOLIDWORKS 2006 SP2, Revision Number 14.2

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetViewPlaneDistance Method (IModelView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : GetViewPlaneDistance Method (IModelView)

Gets the model view plane distance for perspective viewing.

.NET Syntax

Visual Basic (Declaration)

Function GetViewPlaneDistance() As System.Double

Visual Basic (Usage)



```
Dim instance As IModelView
Dim value As System.Double

value = instance.GetViewPlaneDistance()
```

C#

```
System.double GetViewPlaneDistance()
```

C++/CLI

```
System.double GetViewPlaneDistance();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Distance from the eye point to the view plane

Visual Basic for Applications (VBA) Syntax

See [ModelView::GetViewPlaneDistance](#).

Remarks

The value returned is in pixels and represents the distance from the eye point to the view plane (the front of the object). The view plane distance is affected by the view scale.

See Also

[IModelView Interface](#)
[IModelView Members](#)
[IModelView::AddPerspective Method](#)
[IModelView::GetEyePoint Method](#)
[IModelView::GetIsoPlaneDistance Method](#)
[IModelView::HasPerspective Method](#)
[IModelView::IGetEyePoint Method](#)
[IModelView::ISetEyePoint Method](#)
[IModelView::SetEyePoint Method](#)
[IModelView::RemovePerspective Method](#)
[IModelView::Scale2 Property](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetVisibleBox Method (IModelView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : GetVisibleBox Method (IModelView)

Gets the boundaries of the visible model view window.

.NET Syntax

Visual Basic (Declaration)

```
Function GetVisibleBox() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelView
Dim value As System.Object

value = instance.GetVisibleBox()
```

C#

```
System.object GetVisibleBox()
```

C++/CLI

```
System.Object^ GetVisibleBox();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of four longs (win_x_min, win_y_min, win_x_max, and win_y_max) in screen pixels indicating the boundaries of the visible model view window; any part of the model view window blocked by the FeatureManager design tree is excluded

Visual Basic for Applications (VBA) Syntax

See [ModelView::GetVisibleBox](#).

See Also

[IModelView Interface](#)
[IModelView Members](#)
[IModelView::IGetVisibleBox Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GraphicsRedraw Method (IModelView)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : GraphicsRedraw Method (IModelView)

Redraws the specified region of or the entire SOLIDWORKS graphics window.

.NET Syntax

Visual Basic (Declaration)

```
Sub GraphicsRedraw( _  
    ByVal Rect As System.Object _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelView  
Dim Rect As System.Object  
  
instance.GraphicsRedraw(Rect)
```

C#

```
void GraphicsRedraw(  
    System.object Rect  
)
```

C++/CLI

```
void GraphicsRedraw(  
& System.Object^ Rect  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Rect

Array of four integers or longs indicating the boundary of the region in the SOLIDWORKS graphics window to redraw; if the array is empty, then the entire graphics window is redrawn

Visual Basic for Applications (VBA) Syntax

See [ModelView::GraphicsRedraw](#).

Example

[Exclude Faces Before Flattening \(C#\)](#)
[Exclude Faces Before Flattening \(VB.NET\)](#)
[Exclude Faces Before Flattening \(VBA\)](#)

Remarks

This method forces the specified region or the entire SOLIDWORKS graphics window to be updated immediately when called.

See Also

[IModelView Interface](#)
[IModelView Members](#)
[IModelDoc2::WindowRedraw Method](#)
[IModelView::DrawHighlightedItems Method](#)
[IModelView::DrawTransparentFeatureTree Method](#)
[IModelView::SuppressWaitCursorDuringRedraw Property](#)
[IModelView::IGraphicsRedraw Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

HasPerspective Method (IModelView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : HasPerspective Method (IModelView)

Determines if the model view currently has perspective.

.NET Syntax

Visual Basic (Declaration)

```
Function HasPerspective() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelView  
Dim value As System.Boolean  
  
value = instance.HasPerspective()
```

C#

```
System.bool HasPerspective()
```

C++/CLI

```
System.bool HasPerspective();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if the view currently has perspective viewing turned on, false otherwise

Visual Basic for Applications (VBA) Syntax

See [ModelView::HasPerspective](#).

See Also

[IModelView Interface](#)
[IModelView Members](#)
[IModelView::AddPerspective Method](#)
[IModelView::GetEyePoint Method](#)
[IModelView::GetIsoPlaneDistance Method](#)
[IModelView::GetViewPlaneDistance Method](#)

[IModelView::!GetEyePoint Method](#)
[IModelView::!SetEyePoint Method](#)
[IModelView::SetEyePoint Method](#)
[IModelView::RemovePerspective Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

HideMagnifyingGlass Method (IModelView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : HideMagnifyingGlass Method (IModelView)

Hides the Magnifying Glass tool.

.NET Syntax

Visual Basic (Declaration)

```
Sub HideMagnifyingGlass()
```

Visual Basic (Usage)

```
Dim instance As IModelView  
instance.HideMagnifyingGlass()
```

C#

```
void HideMagnifyingGlass()
```

C++/CLI

```
void HideMagnifyingGlass();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelView.HideMagnifyingGlass](#).

Example

Visual Basic for Applications (VBA)

```
'-----  
'  
' Preconditions: Model document is open.  
'  
' Postconditions: None  
'  
'-----  
Option Explicit  
  
Dim swApp As SldWorks.SldWorks  
Dim swModel As SldWorks.ModelDoc2  
Dim swModelView As SldWorks.ModelView  
Sub main()  
  
Set swApp = Application.SldWorks  
Set swModel = swApp.ActiveDoc  
Set swModelView = swModel.ActiveView  
  
swModelView.ShowMagnifyingGlass -0.01928933522023, 0.004431675106825,  
-0.001816629754713, 2, True, True  
swModelView.MoveMagnifyingGlass -0.01928933522023, 0.004431675106825, -0.004  
swModelView.MoveMagnifyingGlass -0.01928933522023, 0.004431675106825, -0.016  
swModelView.HideMagnifyingGlass  
  
End Sub
```

See Also

[IModelView Interface](#)
[IModelView Members](#)
[IModelView::MoveMagnifyingGlass Method](#)
[IModelView::ShowMagnifyingGlass Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

HlrQuality Property (IModelView)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : HlrQuality Property (IModelView)

Gets or sets the hidden-line removal quality property of this model view.

.NET Syntax

Visual Basic (Declaration)

```
Property HlrQuality As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModelView  
Dim value As System.Integer  
  
instance.HlrQuality = value  
  
value = instance.HlrQuality
```

C#

```
System.int HlrQuality {get; set;}
```

C++/CLI

```
property System.int HlrQuality {  
    System.int get();  
    void set ( & System.int value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Type of HLR done as defined in [swHlrQuality_e](#)

Visual Basic for Applications (VBA) Syntax

See [ModelView::HlrQuality](#).

Example

[Get and Set Model View HLR Quality \(VBA\)](#)
[Get and Set Model View HLR Quality \(VB.NET\)](#)
[Get and Set Model View HLR Quality \(C#\)](#)

See Also

[IModelView Interface](#)
[IModelView Members](#)
[IModelView::DynamicMode Property](#)
[IModelView::DisplayMode Property](#)
[IModelView::StopDynamics Method](#)
[IModelView::StartDynamics Method](#)

Availability

SOLIDWORKS 2000 FCS, Revision Number 8.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetEyePoint Method (IModelView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : IGetEyePoint Method (IModelView)

Gets the eye position for perspective viewing.

.NET Syntax

Visual Basic (Declaration)

```
Sub IGetEyePoint(  
    ByRef Eyept As System.Double _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelView  
Dim Eyept As System.Double  
  
instance.IGetEyePoint(Eyept)
```

C#

```
void IGetEyePoint(  
    ref System.double Eyept  
)
```

C++/CLI

```
void IGetEyePoint(  
& System.double% Eyept  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Eyept

Array of 3 doubles describing the eye position in screen space

Visual Basic for Applications (VBA) Syntax

See [ModelView::IGetEyePoint](#).

Remarks

The values are returned in pixels and are the location of the eyepoint relative to the screen space origin (upper-left corner of window). The Z value reflects the distance from the eye point to the object center. These values are affected by the view scale.

To use these values, you should apply them after you have obtained the current view orientation matrix using ModelView::Xform.

The Z value (in pixels) will typically be close to

[([IModelView::GetViewPlaneDistance](#) / ("Object Sizes Away")) * ("Object Sizes Away" + 0.5)]

The perspective view is created based on the value entered by the end-user for **Object Sizes Away**. If the user specifies **3 objects away**, then the eye point is positioned 3 body diameters from the view plane, where the view plane is located at the front of the body as seen from this particular orientation.

See Also

[IModelView Interface](#)
[IModelView Members](#)
[IModelView::AddPerspective Method](#)
[IModelView::GetEyePoint Method](#)
[IModelView::GetIsoPlaneDistance Method](#)

[IModelView::GetViewPlaneDistance Method](#)
[IModelView::HasPerspective Method](#)
[IModelView::ISetEyePoint Method](#)
[IModelView::SetEyePoint Method](#)
[IModelView::RemovePerspective Method](#)
[IModelView::Scale2 Property](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetNext Method (IModelView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : IGetNext Method (IModelView)

Gets the next model view after this model view.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetNext() As ModelView
```

Visual Basic (Usage)



```
Dim instance As IModelView  
Dim value As ModelView  
  
value = instance.IGetNext()
```

C#

```
ModelView IGetNext()
```

C++/CLI

```
ModelView^ IGetNext();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Next [model view](#) or NULL if no more model views exist

Visual Basic for Applications (VBA) Syntax

See [ModelView::IGetNext](#).

Remarks

You can traverse through the model views in a document by using this method to get the initial view and each of the following views. When this method returns Nothing, you have reached the end of the list.

See [IModelDoc2::EnumModelViews](#) for a method for traversing the model views on a document.

See Also

[IModelView Interface](#)
[IModelView Members](#)
[IModelView::GetNext Method](#)
[IModelDoc2::GetFirstModelView Method](#)
[IModelDoc2::IGetFirstModelView Method](#)
[IModelDoc2::ActiveView Property](#)
[IModelDoc2::IActiveView Property](#)
[IEnumModelViews::Next Method](#)

Availability

SOLIDWORKS 98Plus, datecode 1998319

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetStereoSeparation Method (IModelView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : IGetStereoSeparation Method (IModelView)

 .NET Syntax

Visual Basic (Declaration)

```
Function IGetStereoSeparation() As System.Double
```

Visual Basic (Usage)  Copy Code

```
Dim instance As IModelView
Dim value As System.Double

value = instance.IGetStereoSeparation()
```

C#

```
System.double IGetStereoSeparation()
```

C++/CLI

```
System.double IGetStereoSeparation();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value
Array of 2 doubles representing the stereo magnitude and stereo parallax balance settings

 Visual Basic for Applications (VBA) Syntax

See [ModelView::IGetStereoSeparation](#).

 Remarks

This method checks the two stereoscopic display values which can be set by using [IModelView::SetStereoSeparation](#) or [IModelView::ISetStereoSeparation](#).

 See Also

[IModelView Interface](#)
[IModelView Members](#)

IGetVisibleBox Method (IModelView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : IGetVisibleBox Method (IModelView)

Gets the boundaries of the visible model view window.

 .NET Syntax

Visual Basic (Declaration)

```
Function IGetVisibleBox() As System.Integer
```

Visual Basic (Usage)  Copy Code

```
Dim instance As IModelView
Dim value As System.Integer

value = instance.IGetVisibleBox()
```

C#

```
System.int IGetVisibleBox()
```

C++/CLI

```
System.int IGetVisibleBox();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of four longs (win_x_min, win_y_min, win_x_max, and win_y_max) in screen pixels indicating the boundaries of the visible model view window; any part of the model view window blocked by the FeatureManager design tree is excluded

Visual Basic for Applications (VBA) Syntax

See [ModelView::IGetVisibleBox](#).

See Also

[IModelView Interface](#)
[IModelView Members](#)
[IModelView::GetVisibleBox Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGraphicsRedraw Method (IModelView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : IGraphicsRedraw Method (IModelView)

Redraws the specified region of or the entire SOLIDWORKS graphics window.

.NET Syntax

Visual Basic (Declaration)

```
Sub IGraphicsRedraw( _  
    ByRef Rect As System.Integer _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelView  
Dim Rect As System.Integer  
  
instance.IGraphicsRedraw(Rect)
```

C#

```
void IGraphicsRedraw(  
    ref System.int Rect  
)
```

C++/CLI

```
void IGraphicsRedraw(  
& System.int% Rect  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Rect

Array of four integers or longs indicating the boundary of the region in the SOLIDWORKS graphics window to redraw; if the array is empty, then the entire graphics window is redrawn

Visual Basic for Applications (VBA) Syntax

See [ModelView::IGraphicsRedraw](#).

Remarks

This method forces the specified region or the entire SOLIDWORKS graphics window to be updated immediately when called.

See Also

[IModelView Interface](#)
[IModelView Members](#)
[IModelDoc2::WindowRedraw Method](#)
[IModelView::DrawHighlightedItems Method](#)
[IModelView::DrawTransparentFeatureTree Method](#)
[IModelView::GraphicsRedraw Method](#)
[IModelView::SuppressWaitCursorDuringRedraw Property](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InitializeShading Method (IModelView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : InitializeShading Method (IModelView)

Sets up the model view for OpenGL shading.

.NET Syntax

Visual Basic (Declaration)

```
Sub InitializeShading()
```

Visual Basic (Usage)



```
Dim instance As IModelView  
instance.InitializeShading()
```

C#

```
void InitializeShading()
```

C++/CLI

```
void InitializeShading();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelView::InitializeShading](#).

Remarks

Third-party developers should call this method if they are going to use OpenGL shading and are unsure whether shading has already been used for the current window. This method needs to be called for any window drawn into with OpenGL calls.

This method is provided so third-party applications do not need to use any of the Choos...() functions provided in the OpenGL library. Many of the Choos...() functions, for example ChoosePixelFormat(), can only be called once per window. In the case of ChoosePixelFormat(), a second call to the function is ignored. The effect is harmless if the third-party application makes this call after the SOLIDWORKS Design application has made the call. However, it is dangerous if the third party were to call it first because the SOLIDWORKS Design application may require a more complicated format.

To make OpenGL calls, you must get an OpenGL render context (RC). We keep our render context private so that there are not any problems with corruption of SOLIDWORKS RC by an application. To get a rendering context to make OpenGL calls, you must:

```
uiModelView_c::InitializeShading()  
CClientDC dC(CWnd* parentWindow);  
HGLRC hRc = ::wglCreateContext(dC.m_hDC);
```

```
::wglMakeCurrent(dc.m_hDC, hRc); (call this before each 'paint' operation)
```

When you are done using OpenGL for this paint operation, call:

```
::wglMakeCurrent(NULL, NULL);
```

When the DLL shuts down or the [IModelView](#) object is destroyed (watch for a [DModelViewEvents_DestroyNotify2](#)), your application must do the following:

```
::wglMakeCurrent(NULL, NULL);
```

```
::wglDeleteContext(hRc);
```

See Also

[IModelView Interface](#)
[IModelView Members](#)
[IModelDoc2::ViewOglShading Method](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IOrientation Property (IModelView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : IOrientation Property (IModelView)

Obsolete. Superseded by [IModelView::Orientation3](#).

.NET Syntax

Visual Basic (Declaration)

```
Property IOrientation As System.Double
```

Visual Basic (Usage)



```
Dim instance As IModelView  
Dim value As System.Double  
  
instance.IOrientation = value  
  
value = instance.IOrientation
```

C#

```
System.double IOrientation {get; set;}
```

C++/CLI

```
property System.double IOrientation {  
    System.double get();  
    void set ( & System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelView::IOrientation](#).

See Also

[IModelView Interface](#)
[IModelView Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IOrientation2 Property (IModelView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : IOrientation2 Property (IModelView)

Obsolete. Superseded by [IModelView::Orientation3](#).

.NET Syntax

Visual Basic (Declaration)

```
Property IOrientation2 As System.Double
```

Visual Basic (Usage)



```
Dim instance As IModelView  
Dim value As System.Double  
  
instance.IOrientation2 = value  
value = instance.IOrientation2
```

C#

```
System.double IOrientation2 {get; set;}
```

C++/CLI

```
property System.double IOrientation2 {  
    System.double get();  
    void set ( & System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelView::IOrientation2](#).

See Also

[IModelView Interface](#)
[IModelView Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

ISetEyePoint Method (IModelView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : ISetEyePoint Method (IModelView)

Sets the eye position for perspective viewing.

.NET Syntax

Visual Basic (Declaration)

```
Function ISetEyePoint( _  
    ByRef Eyept As System.Double _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelView  
Dim Eyept As System.Double  
Dim value As System.Boolean  
  
value = instance.ISetEyePoint(Eyept)
```

C#

```
System.bool ISetEyePoint(  
    ref System.double Eyept  
)
```

C++/CLI

```
System.bool ISetEyePoint(  
& System.double% Eyept  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Eyept
Array of 3 doubles describing the eye position in screen space

Return Value

True if the eye point is set successfully, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelView::ISetEyePoint](#).

Remarks

The values are returned in pixels and are the location of the eye point relative to the screen-space origin (upper-left corner of window). The Z value, in pixels, reflects the distance from the eye point to the object center. These values will be affected by the view scale.

To use these values, you should apply them after you have obtained the current view orientation matrix using [IModelView::Transform](#).

The Z value typically are close to

[([IModelView::GetViewPlaneDistance](#) / ("Object Sizes Away")) * ("Object Sizes Away" + 0.5)]

See Also

[IModelView Interface](#)
[IModelView Members](#)
[IModelView::AddPerspective Method](#)
[IModelView::GetEyePoint Method](#)
[IModelView::GetIsoPlaneDistance Method](#)
[IModelView::GetViewPlaneDistance Method](#)
[IModelView::HasPerspective Method](#)
[IModelView::IGetEyePoint Method](#)
[IModelView::SetEyePoint Method](#)
[IModelView::RemovePerspective Method](#)
[IModelView::Scale2 Property](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

ISetStereoSeparation Method (IModelView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : ISetStereoSeparation Method (IModelView)

Obsolete and not superseded. Functionality no longer implemented.

.NET Syntax

Visual Basic (Declaration)

```
Function ISetStereoSeparation( _  
    ByRef DfSeparation As System.Double _  
) As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As IModelView  
Dim DfSeparation As System.Double  
Dim value As System.Boolean
```

```
value = instance.ISetStereoSeparation(DfSeparation)
```

C#

```
System.bool ISetStereoSeparation(  
    ref System.double DfSeparation  
)
```



```
System.bool ISetStereoSeparation(
& System.double% DfSeparation
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

DfSeparation

Array of 2 doubles representing the stereo magnitude and stereo parallax balance settings

Return Value

True if changing the stereo-value settings is successful, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelView::ISetStereoSeparation](#).

Remarks

This method only affects the model view's display if:

- Display window was set up to display stereoscopically
- Application is currently set to display stereoscopically.

[ISldWorks::EnableStereoDisplay](#) needs to be called to do these things.

There are two stereoscopic display values this method sets.

1. Settings[0] controls the magnitude of the stereoscopic effect by specifying the stereoscopic, virtual, camera separation. Appropriate values for this setting depend on the scale of the object, the distance from the virtual camera to the object center, the perspective field of view angle, and, perhaps, an adjustment factor that is available to the user.
2. Settings[1] controls the asymmetry of the projection frustums, thus, adjusting the parallax balance of the stereoscopic scene. It is desirable for any stereoscopic scene to comfortably balance the use of negative parallax (elements appearing to float in front of the display surface) and positive parallax (elements appearing to reside somewhere behind the display surface). Appropriate values for this setting should be arrived at experimentally.

If this method is not called, the default settings will equal 0.0, resulting in identical left-eye and right-eye renderings, and no visible stereo effect, even if stereoscopic display is currently enabled.

See Also

[IModelView Interface](#)
[IModelView Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

ITranslation Property (IModelView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : ITranslation Property (IModelView)

Obsolete. Superseded by [IModelView::Translation3](#).

.NET Syntax

Visual Basic (Declaration)

```
Property ITranslation As System.Double
```

Visual Basic (Usage)



```
Dim instance As IModelView
Dim value As System.Double

instance.ITranslation = value

value = instance.ITranslation
```

C#

```
System.double ITranslation {get; set;}
```

C++/CLI

```
property System.double ITranslation {  
    System.double get();  
    void set ( &    System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelView::ITranslation](#).

See Also

[IModelView Interface](#)
[IModelView Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

ITranslation2 Property (IModelView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : ITranslation2 Property (IModelView)

Obsolete. Superseded by [IModelView::Translation3](#).

.NET Syntax

Visual Basic (Declaration)

```
Property ITranslation2 As System.Double
```

Visual Basic (Usage)



```
Dim instance As IModelView  
Dim value As System.Double  
  
instance.ITranslation2 = value  
  
value = instance.ITranslation2
```

C#

```
System.double ITranslation2 {get; set;}
```

C++/CLI

```
property System.double ITranslation2 {  
    System.double get();  
    void set ( &    System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelView::ITranslation2](#).

See Also

[IModelView Interface](#)
[IModelView Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

Linked Property (IModelView)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : Linked Property (IModelView)

Gets whether or not this viewport is linked or not.

.NET Syntax

Visual Basic (Declaration)

ReadOnly Property Linked As System.Boolean

Visual Basic (Usage)



```
Dim instance As IModelView
Dim value As System.Boolean
```

```
value = instance.Linked
```

C#

```
System.bool Linked {get;}
```

C++/CLI

```
property System.bool Linked {
    System.bool get();
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True if this viewport is linked, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelView::Linked](#).

Example

[Set Viewports \(VB.NET\)](#)
[Set Viewports \(VBA\)](#)
[Set Viewports \(C#\)](#)

See Also

[IModelView Interface](#)
[IModelView Members](#)
[IModelViewManager::LinkedViews Property](#)
[IModelViewManager::ViewportDisplay Property](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

MoveMagnifyingGlass Method (IModelView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : MoveMagnifyingGlass Method (IModelView)

Moves Magnifying Glass tool to the specified coordinates.

.NET Syntax

Visual Basic (Declaration)

```
Sub MoveMagnifyingGlass( _
    ByVal PtX As System.Double, _
    ByVal PtY As System.Double, _
```

```
ByVal Ptz As System.Double _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelView  
Dim Ptx As System.Double  
Dim Pty As System.Double  
Dim Ptz As System.Double  
  
instance.MoveMagnifyingGlass(Ptx, Pty, Ptz)
```

C#

```
void MoveMagnifyingGlass(  
    System.double Ptx,  
    System.double Pty,  
    System.double Ptz  
)
```

C++/CLI

```
void MoveMagnifyingGlass(  
& System.double Ptx,  
& System.double Pty,  
& System.double Ptz  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Ptx
x coordinate
Pty
y coordinate
Ptz
z coordinate

Visual Basic for Applications (VBA) Syntax

See [ModelView.MoveMagnifyingGlass](#).

Example

Visual Basic for Applications (VBA)

```
'-----  
'  
' Preconditions: Model document is open.  
'  
' Postconditions: None  
'  
'-----  
Option Explicit  
  
Dim swApp As SldWorks.SldWorks  
Dim swModel As SldWorks.ModelDoc2  
Dim swModelView As SldWorks.ModelView  
Sub main()  
  
Set swApp = Application.SldWorks  
Set swModel = swApp.ActiveDoc  
Set swModelView = swModel.ActiveView  
  
swModelView.ShowMagnifyingGlass -0.01928933522023, 0.004431675106825,  
-0.001816629754713, 2, True, True  
swModelView.MoveMagnifyingGlass -0.01928933522023, 0.004431675106825, -0.004  
swModelView.MoveMagnifyingGlass -0.01928933522023, 0.004431675106825, -0.016  
swModelView.HideMagnifyingGlass  
  
End Sub
```

See Also

[IModelView Interface](#)
[IModelView Members](#)
[IModelView::HideMagnifyingGlass Method](#)
[IModelView::ShowMagnifyingGlass Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

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ObjectSizesAway Property (IModelView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : ObjectSizesAway Property (IModelView)

Helps define the perspective of the current model view by relating the size of a displayed object with the distance of the object from the observer.

.NET Syntax

Visual Basic (Declaration)

Property ObjectSizesAway As System.Double

Visual Basic (Usage)



```
Dim instance As IModelView
Dim value As System.Double

instance.ObjectSizesAway = value
value = instance.ObjectSizesAway
```

C#

```
System.double ObjectSizesAway {get; set;}
```

C++/CLI

```
property System.double ObjectSizesAway {
    System.double get();
    void set ( & System.double value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Distance of the object from the observer, relative to the size of the object

Visual Basic for Applications (VBA) Syntax

See [ModelView::ObjectSizesAway](#).

Remarks

This property controls the same value as the view, display, perspective information dialog box. It gives the ratio of the distance of the object from the observer to the size of the object. The smaller the value, the greater the amount of perspective distortion.

You can only modify or get this property when the current model view has the perspective display enabled. See [IModelView::AddPerspective](#) and [IModelView::RemovePerspective](#). If perspective display is disabled, getting the property returns -1, and setting the property has no effect.

See Also

[IModelView Interface](#)
[IModelView Members](#)
[IModelView::HasPerspective Method](#)

Orientation Property (IModelView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : Orientation Property (IModelView)

Obsolete. Superseded by [IModelView::Orientation3](#).

.NET Syntax

Visual Basic (Declaration)

Property Orientation As System.Object

Visual Basic (Usage)



```
Dim instance As IModelView  
Dim value As System.Object
```

```
instance.Orientation = value
```

```
value = instance.Orientation
```

C#

```
System.object Orientation {get; set;}
```

C++/CLI

```
property System.Object^ Orientation {  
    System.Object^ get();  
    void set ( & System.Object^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelView::Orientation](#).

See Also

[IModelView Interface](#)
[IModelView Members](#)

Orientation2 Property (IModelView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : Orientation2 Property (IModelView)

Obsolete. Superseded by [IModelView::Orientation3](#).

.NET Syntax

Visual Basic (Declaration)

Property Orientation2 As System.Object

Visual Basic (Usage)



```
Dim instance As IModelView  
Dim value As System.Object
```

```
instance.Orientation2 = value
```

```
value = instance.Orientation2
```

C#

```
System.Object Orientation2 {get; set;}
```

C++/CLI

```
property System.Object^ Orientation2 {  
    System.Object^ get();  
    void set ( & System.Object^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelView::Orientation2](#).

See Also

[IModelView Interface](#)
[IModelView Members](#)

SOLIDWORKS API Help

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Orientation3 Property (IModelView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : Orientation3 Property (IModelView)

Gets or sets the model view orientation matrix.

.NET Syntax

Visual Basic (Declaration)

```
Property Orientation3 As MathTransform
```

Visual Basic (Usage)



```
Dim instance As IModelView  
Dim value As MathTransform  
  
instance.Orientation3 = value  
  
value = instance.Orientation3
```

C#

```
MathTransform Orientation3 {get; set;}
```

C++/CLI

```
property MathTransform^ Orientation3 {  
    MathTransform^ get();  
    void set ( & MathTransform^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

[View orientation matrix](#)

Visual Basic for Applications (VBA) Syntax

See [ModelView::Orientation3](#).

Remarks

When modifying a model view transform, you must use [IModelView::Orientation3](#) and [IModelView::Translation3](#). For example:

...

```

Dim xyzOrigin As SldWorks.MathPoint
dPoint(0) = 0#: dPoint(1) = 0#: dPoint(2) = 0#
Set xyzOrigin = swMathUtil.CreatePoint((dPoint))

Set swVectorZ = swVectorZ.Scale(-1)
Set swMathTrans = swMathUtil.ComposeTransform(swVectorX, swVectorY, swVectorZ,
xyzOrigin.ConvertToVector, 1#)
Set swMathTrans = swMathTrans.Inverse

Set swModelView = swModel.ActiveView
Set swViewTrans = swModelView.Orientation3
swModelView.Orientation3 = swMathTrans

Dim u As Double
u = swModelView.Scale2
Set swMathPoint = swVectorT.ConvertToPoint
Set swMathPoint = swMathPoint.Scale(-1 * u)
Set swMathPoint = swMathPoint.MultiplyTransform(swMathTrans)
swModelView.Translation3 = swMathPoint.ConvertToVector
...

```

Also, check the vectors by looking at the triad in the model view.

See Also

[IModelView Interface](#)
[IModelView Members](#)
[IModelView::Scale2 Property](#)

Availability

SOLIDWORKS 2001Plus SP2, Revision Number 10.2